ZS-F30
Twin-Reuleaux Reality Structure and the Steiner-Centered Support-Ratio Rapidity

A conditional constant-width → Lorentzian causal-cone construction: a translation-invariant rapidity from convex asymmetry, a rapidity-composition generation of SL(2,ℂ) that closes the noncompact gate, and a spinor causal-cone reconstruction — with time reversal and the present kept to a scoped discussion

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Verification (v1.4): SYMBOLIC-THEOREM 4 · GEOMETRY 5 · FULL-HOLONOMY 5 · CAUSAL-CONE 4 · DYNAMICS-DIAGNOSTIC 3 · REGRESSION 5 · SCOPE-AUDIT 5, all consistent | Zero Free Parameters | G-6 CLOSED; open gates registered (§G) | script: zs_f30_verify_v1_4.py

Table 0. Verification ledger, restructured around the construction. No merged PASS count; checks grouped by what they establish. (Pipe characters inside cells are written "mod" to keep the table well-formed.)

Category	Count	Content
SYMBOLIC-THEOREM	4	Exact (SymPy) identities: the Lorentz algebra [K,K] = −J; det X = Minkowski form on $Herm_2(\mathbb{C})$; constant-width ⇔ even cosine **and** sine Fourier modes vanish; the Doubled-Simple Exchange τ-form fixed algebra = so(3,1).
GEOMETRY	5	Support-ratio rapidity dimensionless and seam-odd; a centered circle carries no noncompact excitation; convexity and the seam projectors $P\pm = ½(I\pm J)$; **Steiner-centering gives translation-gauge invariance**; **$\rho = \operatorname{artanh} \beta$** with $\beta=(h_1-h_2)/w$, and the **exact Reuleaux** harmonics (3,9,15,…) vs the corpus leading model.
FULL-HOLONOMY	5	The **actual** generators X(θ,n) have real rank 6 = so(3,1); det G_L = 1 with unitarity iff ρ=0; seam = boost inversion $B(\theta+\pi)=B(\theta)^{-1}$; the pair lies in Fix(σ_Z) (consistency); the **Rapidity-Composition Theorem** generating SL(2,ℂ).

Category	Count	Content
CAUSAL-CONE	4	Null boundary $\partial C_+ = \{\psi\psi^\dagger\}$; interior = future timelike cone; SL(2,ℂ) preserves the future cone and is orthochronous; the η-free signature-(1,3) chain.
DYNAMICS-DIAGNOSTIC	3	Local/numerical diagnostics, not theorems: f_+ vs f_- multipliers; arrow existence (mod $\lambda \neq 1$) vs direction (mod $\lambda < 1$); $T^2=-1$ Kramers value for j=½.
REGRESSION	5	Locked-input re-confirmation: A=35/437, Q=11, (Z,X,Y)=(2,3,6); the four real forms; M2/M17.5 compatibility. Prove nothing new.
SCOPE-AUDIT	5	**G-6 now CLOSED** (composition); the support-ratio↔physical-boost identification is DERIVED-CONDITIONAL (G-7); physical-T OPEN (G-2); no canonical temporal POVM (G-3); "circle SELECTS so(4)" and "Reuleaux MAXIMIZES the rapidity norm" remain OPEN (G-8); no advance on A29, no phenomenology.

§0. Abstract

Why is macroscopic spacetime Lorentzian, signature (1,3), with a determinate light cone and time orientation? In the Z-Spin framework the relevant algebra is the complexified Z-Spin holonomy so(4,ℂ) ≅ sl(2,ℂ) ⊕ sl(2,ℂ) (ZS-M2), and the corpus records (ZS-M17.5) that the Z_2 seam involution $\hat{S}^2=I$ selects the Lorentzian over the Euclidean form by exchanging the two su(2) factors. This paper tightens that selection into a single chain and — repairing v1.3 — supplies the noncompact (boost) direction from convex geometry in a **translation-gauge-invariant** way.

The earlier draft (v1.3) introduced a support-ratio rapidity $\rho = \frac{1}{2}\ln(h_1/h_2)$ but defined it from a raw support function, which depends on the choice of origin ($h_{K+a}(\theta) = h_K(\theta) + a\cdot n(\theta)$): an off-center circle would acquire a spurious boost. v1.4 fixes this by **centering at the Steiner point** s(K) — the canonical translation-covariant center, which in 2D is the n=1 Fourier mode of h. The Steiner-centered rapidity $\rho_K(\theta) = \frac{1}{2}\ln(\tilde{h}_K(\theta) / \tilde{h}_K(\theta+\pi))$, with $\tilde{h}_K(\theta) = h_K(\theta) - s(K)\cdot n(\theta)$, is translation-invariant, and for a constant-width curve (no even modes) with the Steiner mode removed it depends only on the odd harmonics $n \geq 3$. Writing the dimensionless asymmetry $\beta(\theta) = (h_1-h_2)/w = (h_1-h_2)/(h_1+h_2)$, one has $|\beta| < 1$ (subluminal) and **$\rho = \operatorname{artanh} \beta$** exactly — the special-relativistic rapidity–velocity relation — so the log-ratio is not arbitrary but the canonical hyperbolic coordinate of the asymmetry. Under the seam, $\beta \mapsto -\beta$ and $\rho \mapsto -\rho$, exactly as a boost inverts.

The half-angle phase supplies the compact rotation angle φ; together $X_L = \frac{1}{2}(\rho+i\varphi)(n\cdot\sigma)$, with the three axes n from the X-sector so(3), spans the full six-dimensional **so(3,1)** — boost generators $K=\sigma/2$ (Hermitian), rotations $J=-i\sigma/2$ (anti-Hermitian), Lorentzian bracket $[K_i,K_j]=-\varepsilon_{ijk} J_k$. A single Reuleaux curve supplies only a **bounded** rapidity, but we close that gate with a **Rapidity-Composition**

Theorem: since the support-ratio image contains a neighborhood of 0, bounded elementary support-boosts compose ($B(\theta_N)^N$) to any rapidity, and with the SU(2) rotations the KAK factorization generates the whole SL(2,$\mathbb{C}$) identity component. Exponentiating, on Hermitian $X = x_0\mathbb{1}+x\cdot\sigma$, $\det X = x_0^2-|x|^2$ is the Minkowski form; the positive cone $C_+=\{X\geq0\}$ is the future causal cone with null boundary $\partial C_+=\{\psi\psi^\dagger\}$, preserved orthochronously by SL(2,$\mathbb{C}$)→$SO^+(1,3)$.

Seven results. **(F30.1)** the four real forms (regression). **(F30.2)** a **Doubled-Simple Exchange Theorem** in τ-form (σ^2=id automatic) plus a **constant-width Fourier theorem** (cosine and sine even modes vanish). **(F30.3)** the **Steiner-centered support-ratio rapidity** (translation-invariant, ρ=artanh β) and the full six-dimensional lift with the boost-inversion identity. **(F30.4)** the spinor causal-cone reconstruction. **(F30.5)** the assembled chain — explicitly **DERIVED-CONDITIONAL**: the algebraic-representation chain is closed, but its physical identification (ρ = the Z-Spin boost parameter) remains conditional on G-7. **(F30.6)** the two-time exclusion. **(F30.7)** the **Rapidity-Composition Theorem** closing the noncompact gate G-6.

We separate, honestly, three layers of "constant-width curve": the general theorem (any odd harmonics), the **exact Reuleaux triangle** (odd multiples of three, n=3,9,15,…, leading coefficient w/4π, rapidity range ≈ ±0.156), and the corpus **leading-harmonic model** (w/16)cos 3θ (a smooth constant-width curve, range ±0.126); the v1.3 phrase "its entire content is (w/16)cos 3θ" is corrected to apply only to the leading model. We also do not claim that a centered circle *selects* the Euclidean real form (only that it produces no noncompact excitation in this lift), nor that the Reuleaux triangle *maximizes* a rapidity norm (an extremal-geometry question we pose, not solve). Time reversal, the arrow, and the present remain scoped (§9). Zero free parameters.

Epistemic Status Legend

Tag	Meaning
PROVEN	Exact theorem or computation (corpus or standard mathematics; here symbolically verified).
IMPORTED-PROVEN	Theorem from the external literature, imported unchanged.
DERIVED	Valid deduction from PROVEN inputs, no new assumption.
DERIVED-CONDITIONAL	Valid deduction conditional on a stated identification.
DERIVED-interpretation	Structural reading of PROVEN results; not a closed theorem.
HYPOTHESIS-strong	Multiple independent structural anchors; promotion path documented.
NUMERICAL / DIAGNOSTIC	A finite computation or local test; suggestive, not a global theorem.

Tag	Meaning
OPEN	An explicit gate with a stated promotion path (§G).
NON-CLAIM (NC)	Explicitly outside scope.

§1. Introduction

The Z-Spin framework fixes (Z,X,Y) = (2,3,6), **Q** = 11, **A** = 35/437 (ZS-F2, ZS-F5). Granting that the relevant algebra complexifies to $so(4,\mathbb{C}) \cong sl(2,\mathbb{C}) \oplus sl(2,\mathbb{C})$ (ZS-M2, PROVEN), why is the physical real form Lorentzian, so(3,1), with a light cone and a time orientation?

Inside the corpus, ZS-M17.5 (DERIVED-CONDITIONAL) records that the seam involution $\hat{S}^2=I$, by exchanging the two su(2) factors, selects the Lorentzian form (gate F-M17.6 PASS); the J-conjugation $h_2(\theta)=h_1(\theta+\pi)$ of the twin-Reuleaux pair (ZS-F7 §11.2) is its geometric implementation. This paper realizes and tightens that selection. **Outside the corpus**, the real-form classification is classical (Cartan), and the Hermitian-determinant route to the Minkowski form and the light cone is classical (Penrose–Rindler).

This version repairs the v1.3 review's four substantive points and adds two. (1) The support-ratio rapidity of v1.3 was defined from a raw support function and was therefore **origin-dependent** — a translated circle would show a fictitious boost. v1.4 centers at the **Steiner point**, making the rapidity a translation-gauge invariant (§5.2). (2) v1.3 conflated the **exact Reuleaux triangle** with the corpus's smooth $(w/16)\cos 3\theta$ model; v1.4 separates three layers and labels the ±0.126 range as the leading model's, the exact Reuleaux being ±0.156 with harmonics 3,9,15,… (§5.4). (3) v1.3 left full noncompact surjectivity OPEN; v1.4 **closes it** with a Rapidity-Composition Theorem (§7, F30.7). (4) v1.3's title and conclusion over-claimed a closed physical chain; v1.4 marks the chain **DERIVED-CONDITIONAL**, with the residual physical identification isolated as G-7 (§6.5/§7.1, §G). Two further upgrades: the Doubled-Simple Exchange Theorem is recast in a clean **τ-form** for which σ^2=id is automatic (§4.2), and the choice of the log-ratio is justified canonically by **$\rho = \operatorname{artanh} \beta$** (§5.3). Time reversal, the arrow, and the present remain scoped to §9.

§2. Locked inputs and the seam involution as twin-Reuleaux geometry

Dimensional skeleton, no free parameter: dim(Z)=2, dim(X)=3, dim(Y)=6, **Q**=11, **A**=35/437. The complexified holonomy algebra is $so(4,\mathbb{C}) \cong sl(2,\mathbb{C})\_L \oplus sl(2,\mathbb{C})\_R$ (ZS-M2, PROVEN), the two factors commuting.

The seam involution $\hat{S}$ (ZS-F5, PROVEN, $\hat{S}^2=I$) acts on the Z-sector field space as $\theta \to \theta+\pi$. Its geometric realization is the twin-Reuleaux pair (R_1,R_2) of constant width w (ZS-F7 §11.2), with support functions satisfying

$h_2(\theta) = h_1(\theta+\pi) = w - h_1(\theta)$, $R_2 = J(R_1)$, $V\_XZ \propto e^{+i\theta/2}$, $V\_ZY \propto e^{-i\theta/2}$, $\mathbf{V\_ZY = (V\_XZ)^*}$.

Decompose the support function into its seam-even and seam-odd parts (we use h_sym, h_odd to avoid any clash with the twin members h_1,h_2):

$h(\theta) = h\_sym(\theta) + h\_odd(\theta)$, $h\_sym = ½[h(\theta)+h(\theta+\pi)] = w/2$, $h\_odd = ½[h(\theta)-h(\theta+\pi)]$,

so the twin pair is $h_1 = w/2 + u$, $h_2 = w/2 - u$ with u = h_odd. The entire **non-circularity** lives in h_odd. Two facts we use: (i) J carries R_1 to R_2 — it **exchanges** the members (§4); and (ii) the support asymmetry h_odd, suitably normalized and centered, is a **rapidity** (§5). Both rest on the corpus-PROVEN constant-width identity (ZS-F7, J-conjugation to $<10^{-40}$).

§3. Theorem F30.1 — Four-real-form exhaustion (regression)

For **g**_ℂ = sl(2,ℂ) ⊕ sl(2,ℂ) there are exactly four real forms of orthogonal type; their Killing signatures B(X,Y)=Tr(ad_X ad_Y) are:

Table 1. The four real forms of so(4,ℂ) and their Killing signatures (Appendix A; independently recomputed). Signature = (positive, negative).

Defining involution σ	Real form	Structure	Killing signature
factor-preserving, compact	so(4)	su(2) ⊕ su(2)	(0, 6)
factor-preserving, split	so(2,2)	su(1,1) ⊕ su(1,1)	(4, 2)
factor-preserving, mixed	so*(4)	su(2) ⊕ su(1,1)	(2, 4)
factor-EXCHANGING (swap)	so(3,1)	sl(2,ℂ)_ℝ	(3, 3)

The four signatures are pairwise distinct, so a real form is fixed by the **type** of its defining involution. [F30.1: IMPORTED-PROVEN (Cartan); re-run as REGRESSION F4.]

§4. Theorem F30.2 — Exchange-involution fixed algebra, the Doubled-Simple Exchange Theorem (τ-form), and the constant-width Fourier theorem

4.1 The constant-width Fourier theorem. Expand $h(\theta) = w/2 + \Sigma\_\{n\geq 1\}(a\_n \cos n\theta + b\_n \sin n\theta)$.

Theorem F30.2a. The constant-width condition $h(\theta)+h(\theta+\pi) = w$ holds **iff** all even Fourier modes vanish, $a\_\{2k\}=b\_\{2k\}=0$. The seam operator $(Jh)(\theta):=h(\theta+\pi)$ acts as **J = −I on the odd-harmonic subspace**, the projectors are $P\pm = ½(I \pm J)$, and the constant-width deformation space is the J-odd subspace P_-.

Proof. $\cos n(\theta+\pi) = (-1)^n \cos n\theta$ and $\sin n(\theta+\pi) = (-1)^n \sin n\theta$; hence $h(\theta)+h(\theta+\pi) = w + 2\Sigma_{k\geq1}(a_{2k}\cos 2k\theta + b_{2k}\sin 2k\theta)$, equal to w identically iff every even coefficient vanishes. On the odd harmonics $(-1)^n=-1$, so $J=-I$; $P\pm=\frac{1}{2}(I\pm J)$ are complementary. [F30.2a: PROVEN; the cosine **and** sine identities verified symbolically, check A3; convexity $h+h'' \geq 0$ holds for the leading mode, B3.] The twin pair $h\pm = w/2 \pm u_{odd}$ with $u_{odd} \in P_-$ is exchanged by J ($u_{odd} \mapsto -u_{odd}$): **J is factor-exchanging**, forced by constant width.

4.2 The Doubled-Simple Exchange Theorem (τ-form). v1.3 gave a normal form $\sigma_\alpha(X,Y)=(\alpha(\bar{Y}),\alpha^{-1}(\bar{X}))$ for which σ^2=id is **not** automatic for arbitrary α. We use a cleaner form.

Theorem F30.2b. Let **g** be a complex simple Lie algebra and G = **g** ⊕ **g**. Let τ: **g** → **g** be an antilinear Lie-algebra automorphism, and define

$\sigma(X,0) = (0, \tau X)$, $\sigma(0,Y) = (\tau^{-1}Y, 0)$, i.e. $\sigma(X,Y) = (\tau^{-1}Y, \tau X)$.

Then **σ^2 = id automatically** (for any τ), σ is an antilinear involutive automorphism exchanging the two ideals, and Fix(σ) = {(X, τX) : X ∈ **g**} ≅ **g**_ℝ, the realification of **g**, the projection π_1: Fix(σ) → **g**_ℝ being a real Lie-algebra isomorphism. For **g** = sl(2,ℂ) with τ = complex conjugation, Fix(σ) = $\{(X, \bar{X})\}$ ≅ sl(2,ℂ)_ℝ ≅ **so(3,1)** (signature (3,3), Appendix A).

Proof. $\sigma^2(X,Y) = \sigma(\tau^{-1}Y, \tau X) = (\tau^{-1}(\tau X), \tau(\tau^{-1}Y)) = (X,Y)$, so σ^2=id for **any** τ; antilinearity and the automorphism property are inherited from τ. The fixed set is the graph {(X,τX)}; π_1 is a real-linear bijection onto **g** carrying the bracket to itself, hence Fix(σ) ≅ **g**_ℝ. For sl(2,ℂ), **g**_ℝ has real dimension 6 and Killing signature (3,3), i.e. so(3,1). [F30.2b: the general theorem is PROVEN analytically (standard real-form theory); the **sl(2,ℂ) instance** is verified symbolically/numerically — σ^2=id for random τ-inputs and the (3,3) signature, check A4. The code verifies the instance, not arbitrary **g**.] With Z-Spin terminology removed this reads as an independent Lie-theoretic proposition (a factor permutation of the Dynkin/Satake diagram of $A_1 \times A_1$).

4.3 Fixed algebra and exclusion. With the antilinear swap $\sigma_Z(X_L,X_R) = (\bar{X}_R, \bar{X}_L)$ (τ = conjugation), Theorem F30.2b gives Fix(σ_Z) ≅ so(3,1). The three factor-**preserving** forms (so(4), so(2,2), so*(4)) come from involutions fixing each factor; the seam, forced by constant width to **exchange** them (§4.1), **excludes all three** and yields so(3,1) — a forcing resting on two theorems. [F30.2: DERIVED, conditional on §4.1.]

§5. Theorem F30.3 — The Steiner-centered support-ratio rapidity and the full six-dimensional holonomy lift

This section closes the first arrow, **constant-width geometry → rotation + rapidity full holonomy**, in a translation-invariant way.

5.1 The origin-dependence of the raw support ratio. The support function depends on the choice of origin: translating K by a vector a sends $h_K(\theta) \mapsto h_K(\theta) + a\cdot n(\theta)$, with $n(\theta)=(\cos \theta, \sin \theta)$. A raw ratio

$\rho\_raw(\theta) = \frac{1}{2}\ln(h(\theta)/h(\theta+\pi))$ is therefore **not** translation-invariant. The clearest failure is the circle: centered at the origin it has $h \equiv R$ and $\rho\_raw \equiv 0$, but shifted by a it has $h\_a(\theta) = R + a\cdot n(\theta)$, giving $\rho\_raw(\theta) = \frac{1}{2}\ln[(R+a\cdot n)/(R-a\cdot n)] \not\equiv 0$ — a **spurious boost** for a body that should carry none (verified, ρ_raw up to 0.35 for $\|a\|=0.34$). v1.3's "circle ⇒ ρ=0" was thus implicitly a gauge choice.

5.2 Steiner centering (the fix). The Steiner point s(K) is the canonical translation-**covariant** center, $s(K+a)=s(K)+a$; in 2D it is determined by the n=1 Fourier mode alone, $s(K) = (1/\pi)\int_0^{2\pi} h\_K(\theta)\, n(\theta)\, d\theta = (a_1, b_1)$. Define the **Steiner-centered support function** and **rapidity**

$$\tilde{h}\_K(\theta) = h\_K(\theta) - s(K)\cdot n(\theta), \quad \rho\_K(\theta) = \tfrac{1}{2}\ln(\, \tilde{h}\_K(\theta) / \tilde{h}\_K(\theta+\pi)\,).$$

Proposition F30.3a (gauge invariance). ρ_K is translation-gauge invariant: under $K \mapsto K+a$, $s \mapsto s+a$ and $\tilde{h}$ is unchanged, so ρ_K is unchanged. Centering removes the n=1 mode; for a constant-width curve (no even modes) the centered support function carries only odd harmonics $n \geq 3$, and the centered circle has $\tilde{h} \equiv w/2$, $\rho \equiv 0$. [PROVEN; checks B2, B4 — raw ρ gauge-dependent (shifted circle 0.35), Steiner-centered ρ invariant to 10^{-6}; the leading model $(w/16)\cos 3\theta$ is already Steiner-centered (its Steiner point is 0).]

5.3 ρ = artanh β: why the log-ratio. Define the dimensionless **asymmetry** $\beta(\theta) := (\, \tilde{h}_1(\theta) - \tilde{h}_2(\theta)\,) / (\, \tilde{h}_1(\theta) + \tilde{h}_2(\theta)\,) = (\, h_1 - h_2\,) / w$. Then $|\beta| < 1$ (the support functions are positive, so $|h_1-h_2| < h_1+h_2 = w$ — β is "subluminal"), and $h_1/h_2 = (1+\beta)/(1-\beta)$, whence

$$\rho = \tfrac{1}{2}\ln(h_1/h_2) = \tfrac{1}{2}\ln[(1+\beta)/(1-\beta)] = \mathbf{artanh\ \beta},$$

the **special-relativistic rapidity–velocity relation** $\xi = \mathrm{artanh}(v/c)$. The log-ratio is therefore not arbitrary: ρ is the unique additive (rapidity) coordinate of the multiplicatively-composing asymmetry β, with β playing the role of v/c. Under the seam, $\beta(\theta+\pi) = -\beta(\theta)$ immediately, hence $\rho(\theta+\pi) = -\rho(\theta)$. [PROVEN; check B5 — $\rho = \mathrm{artanh}\,\beta$ to 10^{-9}, $|\beta|<1$.] (That β is the **physical** velocity-analog of the Z-Spin holonomy remains G-7; the *mathematical* canonicity of ρ is settled here.)

5.4 Three layers of constant-width curve. We separate, as the structure demands:

(L1) **General constant-width:** $h = w/2 + u\_odd$, $u\_odd \in P_-$ arbitrary. The rapidity $\rho = \mathrm{artanh}[(h_1-h_2)/w]$ is defined for any such curve.

(L2) **Exact Reuleaux triangle:** by C_3 symmetry only harmonics that are multiples of 3 survive, and by constant width only odd ones, so u_odd has harmonics **n = 3, 9, 15, …**; the leading coefficient is **w/(4π) ≈ 0.0796 w** and the Steiner-centered rapidity range is ≈ **±0.156**. The exact curve has corners. [check B5 — FFT of the exact Reuleaux support function: harmonics 3,9,15 present, $c_3 = w/4\pi$, range ±0.156.]

(L3) **Corpus leading-harmonic model:** $u\_odd = (w/16)\cos 3\theta \approx 0.0625\, w\cdot\cos 3\theta$ — a **smooth** constant-width curve (the simplest non-circular one), with rapidity range **±0.126** = $\pm\frac{1}{2}\ln(9/7)$. This is the corpus model (ZS-F7 §6.1) and the object used in the verification.

The v1.3 statement "the Reuleaux triangle's entire content is the seam-odd $(w/16)\cos 3\theta$" is **incorrect for the exact curve** and is hereby restricted to L3. The ± 0.126 range is the **leading-model** value; the exact Reuleaux is ± 0.156. None of the downstream algebra depends on which layer is used.

5.5 The full six-dimensional lift. Combine the boost rapidity $\rho(\theta)$ with the rotation angle φ from the half-angle phase ($\arg V_{XZ} = \theta/2$):

$$X_L(\theta,n) = \tfrac{1}{2} (\rho(\theta) + i \varphi) (n\cdot\sigma) = \rho(\theta)(n\cdot K) + \varphi(n\cdot J), \quad G_L = \exp X_L,$$

with $\mathbf{J} = -(i/2)\sigma$ (anti-Hermitian, compact) and $\mathbf{K} = (1/2)\sigma$ (Hermitian, noncompact). Over the three axes from the X-sector so(3), the six generators close into $sl(2,\mathbb{C})_{\mathbb{R}}$ with $[J_i,J_j]=\varepsilon_{ijk} J_k$, $[J_i,K_j]=\varepsilon_{ijk} K_k$, $\mathbf{[K_i,K_j] = -\varepsilon_{ijk} J_k}$ — the final minus sign being the Lorentzian signature (a compact so(4) would read +). All brackets and Hermiticities are verified **symbolically** (check A1). Crucially — addressing the v1.3 review — we test the **actual** generators $X(\theta,n)$ (with $\rho(\theta)$ and $\varphi(\theta)=\theta/2$ both functions of one θ), not an independent basis: since ρ and φ are linearly independent functions, evaluating at two angles per axis yields **real rank 6** (a single angle gives only rank 3), so the actual holonomy image fills so(3,1) (check C1). $\det G_L = 1$ always, and **G_L is unitary iff $\rho=0$** (check C2).

5.6 Seam = boost inversion (the non-circular diagram). About a fixed axis (boost and rotation commute, both $\propto n\cdot\sigma$), $G_L = B(\theta)R(\varphi)$ with $B = \exp(\tfrac{1}{2}\rho\, n\cdot\sigma)$ (Hermitian boost) and $R = \exp(\tfrac{1}{2}i\varphi\, n\cdot\sigma)$ (unitary rotation).

Proposition F30.3b. Under the geometric seam $\theta \to \theta+\pi$ the boost **inverts**, $B(\theta+\pi) = B(\theta)^{-1}$ (ρ recomputed from the centered support functions); B is Hermitian and, for $\rho\neq 0$, non-unitary, while R is unitary. A Lorentz boost satisfies this inversion law; a rotation does not. The seam thus **distinguishes** the noncompact (boost, seam-odd) from the compact (rotation) direction, certifying that β is a rapidity and not a hidden rotation. [DERIVED; check C3 — the seam image $B(\theta+\pi)$ and the matrix inverse $B(\theta)^{-1}$ are computed independently and agree to 10^{-7}.] For completeness, the pair $(X_L, \bar{X}_L)$ lies in $\mathrm{Fix}(\sigma_Z)=so(3,1)$ for all (θ,n) (check C4, structural/near-definitional). We do **not** assert $J = \sigma_Z$; they are distinct $\mathbb{Z}_2$'s.

5.7 A reading, and what it is not. Proposition F30.3a gives a tempting dichotomy: a centered circle has $\rho\equiv 0$ (no boost) and an asymmetric curve has $\rho\not\equiv 0$ (boosts active). The **safe** statement is: *a centered circle produces no noncompact excitation in this lift*. The **stronger** statement — *a centered circle selects the Euclidean real form so(4)* — is **not** proven here: $\rho=0$ means the boost coordinate vanishes in this parameterization, i.e. the holonomy stays in the rotation subgroup, which does not by itself convert the reality involution σ_Z into a factor-preserving one; that conversion would require a separate involution-transition theorem. Likewise, "the Reuleaux triangle is maximally Lorentzian" has no proof yet: define the **asymmetry functional** $R_\infty(K) = \max_\theta |\rho_K(\theta)|$ (or $R_p(K) = (\int|\rho_K|^p)^{1/p}$); R_∞ increases with the seam-odd amplitude, but **area-minimization (Blaschke–Lebesgue) and R_∞-maximization are different optimizations**. We therefore state only: *the Reuleaux minimizer is a natural extremal candidate for maximal noncompact activation; whether it maximizes a specified rapidity* *norm is OPEN* (§G, G-8). [§5.7: DERIVED-interpretation (the safe half); the two stronger statements are OPEN.]

§6. Theorem F30.4 — Spinor–Minkowski reconstruction and the causal cone (η-free)

This is the third and fourth arrows, **SL(2,ℂ) → Lorentzian causal cone**. Represent a spacetime point by $X = x_0\mathbb{1} + x_1\sigma_1 + x_2\sigma_2 + x_3\sigma_3 \in \mathrm{Herm}_2(\mathbb{C})$; $\det X = x_0^2-x_1^2-x_2^2-x_3^2$ is the Minkowski form (check A2). The eigenvalues are $x_0 \pm |x|$, so:

Theorem F30.4 (causal-cone reconstruction). (a) **Future cone** $C_+ := \{X \geq 0\} = \{x_0 \geq |x|\}$; $\mathrm{tr}\, X = 2x_0 > 0$ fixes the future orientation. (b) **Timelike interior** $X > 0 \Leftrightarrow$ future timelike ($\det X > 0$). (c) **Null boundary** $\partial C_+ = \{X \geq 0, \det X = 0, X \neq 0\}$ = {rank-one positive Hermitian} = $\{\psi\psi^\dagger : \psi \in \mathbb{C}^2\{0\}\}$; every future null ray is a Weyl-spinor outer product. (d) **SL(2,ℂ)** acts by $X \mapsto gXg^\dagger$; $\det g = 1 \Rightarrow \det(gXg^\dagger) = \det X$ (Minkowski form preserved); congruence preserves positive-semidefiniteness (C_+ and ∂C_+ preserved); $\mathrm{tr}(gXg^\dagger) > 0$ for $X \geq 0$ (the map is **orthochronous** — time orientation preserved). This is the 2:1 cover $SL(2,\mathbb{C}) \to SO^+(1,3)$ (Penrose–Rindler).

[F30.4: det X = Minkowski and SL(2,ℂ)-invariance PROVEN/IMPORTED-PROVEN; (a)–(d) IMPORTED-PROVEN (standard spinor causal structure), assembled as the corpus reconstruction; checks D1–D4.] The chain is

swap fixed algebra so(3,1) ⟶ SL(2,ℂ) ↷ $\mathrm{Herm}_2(\mathbb{C})$ ⟶ det X = Minkowski ⟶ (signature (1,3) **and** future cone C_+ with null boundary $\{\psi\psi^\dagger\}$, orthochronously preserved),

every arrow a theorem, the metric **and causal** structure read from the determinant. The dynamical metric (which frame the evolution selects) is OPEN (§G, G-1).

§7. Theorems F30.5 and F30.7 — The conditional chain and the Rapidity-Composition Theorem

7.1 The chain (DERIVED-CONDITIONAL).

Theorem F30.5. Given a constant-width twin pair (h_1,h_2), the half-angle channel V_XZ, and the X-sector axes: (1) the Steiner-centered seam-odd part defines the translation-invariant rapidity $\rho = \mathrm{artanh}[(h_1-h_2)/w]$ (F30.3a, §5.3); (2) $X\_L = \frac{1}{2}(\rho+i\varphi)(n\cdot\sigma)$ spans so(3,1), boosts from the asymmetry, rotations from the phase (F30.3); (3) exponentiation gives SL(2,ℂ) preserving det X (F30.4); (4) C_+ is the future cone with null boundary $\{\psi\psi^\dagger\}$, orthochronous (F30.4). Hence

constant-width geometry ⟶ rotation + rapidity full holonomy ⟶ SL(2,ℂ) ⟶ Lorentzian causal cone,

η-free end to end, the boost direction carried by the (gauge-invariant) support asymmetry.

The algebraic-representation chain is closed; its physical identification remains conditional on G-7 — that the geometric rapidity ρ is the physical boost parameter of the Z-Spin holonomy is an identification (made canonical by $\rho=\operatorname{artanh}\beta$), not a theorem. [F30.5: **DERIVED-CONDITIONAL** on G-7; the algebra and spinor map are PROVEN/IMPORTED-PROVEN, the assembly is DERIVED.] An accurate external title for the result is therefore *a conditional constant-width-to-Lorentzian causal-cone construction*.

7.2 The Rapidity-Composition Theorem (closing G-6). A single Reuleaux curve gives a bounded rapidity (L3: $|\rho| \le \frac{1}{2}\ln(9/7) \approx 0.126$; L2: ≈ 0.156). v1.3 left full noncompact surjectivity OPEN; it is not.

Theorem F30.7 (Rapidity Composition). Suppose the support-ratio image $\{\rho(\theta) : \theta\}$ contains an open neighborhood of 0 (true for any non-circular constant-width curve, since ρ is continuous, seam-odd, and not identically zero, hence takes every value in $[-\rho_{max}, \rho_{max}]$). Then for any target rapidity $\xi \in \mathbb{R}$ and axis n, choosing N with $|\xi/N| \le \rho_{max}$ and θ_N with $\rho(\theta_N) = \xi/N$ (intermediate value theorem), the same-axis boosts commute and

$$B(\theta_N)^N = \exp\left(\tfrac{1}{2} N \rho(\theta_N)\, n\cdot\sigma \right) = \exp\left(\tfrac{1}{2} \xi\, n\cdot\sigma \right),$$

so finite products of elementary support-boosts realize **any** boost. Combined with the SU(2) rotations (X-sector axes), the KAK (Cartan polar) factorization $g = U_1 \exp(\frac{1}{2}\xi\, n\cdot\sigma)\, U_2$, $U_1,U_2 \in SU(2)$, shows that finite products of elementary support-boosts and rotations **generate the whole SL(2,$\mathbb{C}$) identity component**.

Proof. Continuity and the IVT give the elementary solve; same-axis boosts commute because both generators are $\propto n\cdot\sigma$, so $B(\theta_N)^N = \exp(\frac{1}{2} N\rho(\theta_N)\, n\cdot\sigma)$; every $g \in SL(2,\mathbb{C})$ has a polar decomposition $g = U\cdot P$ with $U \in SU(2)$ and $P = \sqrt{g^\dagger g}$ positive-definite Hermitian of determinant 1, i.e. $P = \exp(\frac{1}{2}\xi\, n\cdot\sigma)$; diagonalizing P by an SU(2) rotation gives the KAK form. [F30.7: PROVEN; check C5 — arbitrary boosts to $|\xi|\le 6$ reached by composition (worst 2×10^{-11}), and KAK reconstruction of random SL(2,$\mathbb{C}$) elements to 4×10^{-15}. **G-6 is CLOSED.**]

§8. Position relative to the spacetime-signature literature

(F30.6) the physically extractable statement: *a geometry that supplies a factor-exchanging involution and a nonzero (gauge-invariant) support asymmetry on $sl(2,\mathbb{C})\oplus sl(2,\mathbb{C})$ forces the Lorentzian real form with its future cone, and forbids the split (2,2) two-time structure* (and the Euclidean and quaternionic forms). Tegmark argues from well-posedness/observer stability that only (3,1) is predictive; Bars' two-time physics keeps a (2,2)-type structure with extra gauge symmetry — the case our exchange excludes; van Dam–Ng argue from stability/analyticity. The Z-Spin statement is **algebraic and kinematic**: it locates the signature in the involution type and the convex asymmetry carried by a constant-width curve. The classification is classical (Cartan) and the determinant causal reconstruction is classical (Penrose–Rindler); the contribution is the **geometric realization**, with the new element over v1.3 being a **translation-invariant** noncompact direction derived from convex geometry ($\rho=\operatorname{artanh}\beta$), and the noncompact group fully generated by composition (F30.7). [§8: DERIVED-interpretation; external references IMPORTED.]

§9. Discussion: time reversal, the arrow, and the present (scoped)

9.1 Time reversal and the reality involution. $T = U\_T K$; for spin-½, $T^2 = (-1)^{2j} = -1$ is the forced Kramers value (ZS-M3; ZS-Q14; ZS-A28 $\chi\_Z=-1$). σ_Z (selecting so(3,1)) and physical T play different roles; "the twin-Reuleaux J is physical T" is **OPEN**, pending $T\, H\_L(\theta,n)T^{-1} = H\_R(-\theta,n)$ (§G, G-2). [check E3.]

9.2 The arrow: existence vs direction. i^z has fixed point $z^*=0.4383+0.3606i$, multiplier modulus 0.8915. **Existence** of breaking is modulus $\neq 1$; **direction** (future = contraction) is the stronger modulus < 1; both retained. Bare conjugation maps f_+ to $f_-=(-i)^z$, also contracting. A full arrow theorem is here only a **diagnostic**. [§9.2: DERIVED for the split; the global statement NUMERICAL/DIAGNOSTIC; checks E1, E2.]

9.3 The present is undefined, not impossible. The count n is a clock parameter with no canonical temporal POVM; the v1.0 "no-go" was invalid (geometric measures on $\mathbb{N}$ normalize) and remains retracted. Status **OPEN**: a covariant temporal POVM conjugate to the handshake generator (§G, G-3). [check G4.]

§10. Cross-paper dependency and version-conflict check

Table 2. Dependency and conflict audit. "Same $\hat{S}$" marks realization rather than rediscovery; new rows flag v1.4 additions and version-safety.

Result	Prior status	This paper	Relation / conflict check
selection so(3,1) vs so(4)	M17.5 DERIVED-COND	geometric realization	same seam $\hat{S}$ (F7 §11.2); weak framing; F-M17.6 PASS
$so(4,\mathbb{C})\cong sl(2,\mathbb{C})^2$	M2 PROVEN	used	cited; factors commute (check F5)
four-form exhaustion	F30.1 (v1.2)	regression	unchanged (check F4)
Doubled-Simple Exchange	v1.3 (α-form)	repaired to τ-form	σ^2=id now automatic; same conclusion
constant-width Fourier	F7 §6.1 (example)	PROVEN (general)	cosine **and** sine; no conflict

Result	Prior status	This paper	Relation / conflict check
support-ratio rapidity	v1.3 (raw, gauge-dep)	Steiner-centered, ρ=artanh β	**repairs origin-dependence**; gauge-invariant
exact Reuleaux vs model	v1.3 (conflated)	three layers separated	±0.126 is L3 model; exact L2 is ±0.156; corrects overclaim
holonomy lift	v1.3	actual-generator rank-6	tests the real image, not a basis
noncompact surjectivity	v1.3 OPEN (G-6)	**CLOSED** (F30.7)	composition + KAK
det $\mathrm{Herm}_2(\mathbb{C})$=Minkowski + cone	v1.3	unchanged	causal cone retained
z*, modulus 0.8915	M1 PROVEN	diagnostic	no drift (check E1)

The i-tetration constant z* (ZS-M1) is unmodified; **A** = 35/437, **Q** = 11, dim(Z) = 2 are LOCKED and re-confirmed (F1–F3). Framing is the weaker, accurate one; **no upstream value is modified** — the version-conflict check is clean.

§11. Falsification gates

Table 3. Genuine theorem-falsifiers separated from diagnostics.

Gate	Kind	Trigger
F-F30.1	GENUINE	If any of the four Killing signatures differs from (0,6),(4,2),(2,4),(3,3).
F-F30.2	GENUINE	If Fix(σ_Z) ≠ (3,3), or if constant width did NOT force all even Fourier modes (cosine and sine) to vanish.
F-F30.3a	GENUINE	If the **Steiner-centered** rapidity were not translation-invariant, or if $\rho \neq$ artanh β, or if it were not seam-odd.
F-F30.3b	GENUINE	If $\{J_i,K_i\}$ did not satisfy $[K_i,K_j] = -\varepsilon_{ijk} J_k$, or if the actual generators did not reach real rank 6.
F-F30.3c	GENUINE	If the seam did not invert the boost, $B(\theta+\pi) \neq B(\theta)^{-1}$.
F-F30.4	GENUINE	If det X ≠ Minkowski, if $SL(2,\mathbb{C})$ did not preserve it, if the null boundary were not $\{\psi\psi^\dagger\}$, or if the action were not orthochronous.

Gate	Kind	Trigger
F-F30.7	GENUINE	If bounded elementary support-boosts plus SU(2) rotations did **not** generate SL(2,ℂ) — G-6 would reopen.
F-F30.6	STRUCTURAL-PRED	Observation of stable two-time (2,2) dynamics would contradict the exclusion (§4, §8).
F-F30.8	ANTI-OVERCLAIM	If any step is read as identifying the geometric ρ with the physical boost parameter (G-7), as proving a centered circle *selects* so(4) or the Reuleaux triangle *maximizes* a rapidity norm (G-8), as deriving the dynamical metric (G-1), as identifying σ_Z with physical T (G-2), or as a temporal-Born no-go, it is falsified by this paper's scope (§5–§7, §9, §G).

§G. Open Gates Registry

Gate	Statement	Promotion path
G-1 dynamical metric	Which point/tetrad in $\mathrm{Herm}_2(\mathbb{C})$ the cosmological dynamics selects is open; §6 gives the form and cone, not the frame.	Derive the frame from the Z-sector EOM; prove the flow preserves Fix(σ_Z).
G-2 physical T	σ_Z is a reality involution; its identity with physical T is unproven.	Prove $T\, H_L(\theta,n)\, T^{-1} = H_R(-\theta,n)$.
G-3 temporal POVM	No canonical covariant temporal POVM is assigned to the count n.	Construct a clock observable conjugate to the handshake generator.
G-4 holonomy premises	The lift is conditional on the so(3) axes = SU(2) axes (F18/S15).	Derive the axis identification from F18 first principles.
G-5 A29 G1–G3	Projector selection, flux collectivization, rank-to-energy map (not addressed).	ZS-A30; A29 §8.3.
G-6 noncompact surjectivity	**CLOSED by F30.7** (Rapidity Composition): bounded support-boosts + SU(2) rotations generate SL(2,ℂ).	— (closed).
G-7 physical rapidity	ρ = artanh β is the canonical *mathematical* rapidity of the convex asymmetry; its identity with the *physical* boost parameter of the Z-Spin holonomy is an identification.	Derive ρ as the holonomy of the Z-sector connection along the asymmetry direction, or match it to the F14 joint-ODE boost.

Gate	Statement	Promotion path
G-8 extremality	Whether the centered circle *selects* the Euclidean form (vs merely carrying no noncompact excitation), and whether the Reuleaux triangle *maximizes* $R_\infty(K) = \max_\theta \text{mod-}\rho\_K(\theta)$ (vs minimizing area), are open.	An involution-transition theorem (circle → so(4)); a convex-optimization proof that the Blaschke–Lebesgue minimizer extremizes R_∞.

§12. Conclusion

F30 now stands as seven linked results with the chain made honest. The four real forms are exhausted (F30.1); a τ-form Doubled-Simple Exchange Theorem and a constant-width Fourier theorem make the swap-to-Lorentzian forcing general (F30.2). The boost direction is supplied by the **Steiner-centered** support-ratio rapidity $\rho = \text{artanh}[(h_1-h_2)/w]$ — a **translation-gauge invariant** with the special-relativistic rapidity form, repairing v1.3's origin-dependent definition (F30.3). The actual holonomy generators (not a chosen basis) reach the full six-dimensional so(3,1), the seam acting as boost inversion (F30.3). Exponentiating to $SL(2,\mathbb{C})$ reconstructs the full causal structure — future cone, time orientation, null = $\{\psi\psi^\dagger\}$ — orthochronously (F30.4). A **Rapidity-Composition Theorem** shows bounded elementary support-boosts and SU(2) rotations generate the whole $SL(2,\mathbb{C})$ identity component, **closing the noncompact gate G-6** (F30.7).

The honest status is **a zero-parameter, η-free, DERIVED-CONDITIONAL construction of the Lorentzian causal structure from translation-invariant constant-width asymmetry, with the noncompact group fully generated**. The algebraic-representation chain is closed; the single physical identification (ρ = the Z-Spin boost parameter) is canonical but conditional (G-7), and two interpretive over-readings are held OPEN rather than claimed: a centered circle producing no excitation is not yet the *selection* of so(4), and the area-minimizing Reuleaux triangle is not yet shown to *maximize* the rapidity norm (G-8). The mathematics is self-contained enough to read outside the corpus — its two externally interesting cores are the Doubled-Simple Exchange Theorem and the convex-asymmetry-to-rapidity map $\rho = \text{artanh}\,\beta$ with its composition theorem. The next priorities are the dynamical metric (G-1) and A30 (A29 §8.3), not a further F30 revision.

Acknowledgements & Code Availability

Revised in response to a technical review of v1.3. Repairs and additions: the **Steiner-centered** support-ratio rapidity, removing the origin-dependence of the raw ratio (§5.1–5.2); the **$\rho = \text{artanh}\,\beta$** canonicalization (§5.3); the explicit **three-layer** separation of general constant-width / exact Reuleaux (harmonics 3,9,15, leading w/4π, range ±0.156) / corpus leading model (w/16, range ±0.126) (§5.4), correcting the v1.3 "entire content is (w/16)cos 3θ"; the **Rapidity-Composition Theorem** closing G-6 (§7.2, F30.7); the τ-form **Doubled-Simple Exchange Theorem** with σ^2=id automatic (§4.2); the **actual-generator** rank-6 test (§5.5); the reduction of overclaim — F30.5 marked

DERIVED-CONDITIONAL with the chain's physical step isolated as G-7, and the circle/Reuleaux dichotomy demoted to a safe half plus OPEN G-8 (§5.7). The verification script zs_f30_verify_v1_4.py adds the Steiner translation-invariance test, the ρ=artanh β and exact-Reuleaux-harmonic checks, the sine half of the Fourier identity, the actual-generator rank test, and the composition/KAK checks; it contains no fail-open clause and exits 0 iff every check is consistent.

Appendix A. The four real forms and their Killing signatures

With $\mathbf{g}_\mathbb{C} = sl(2,\mathbb{C})\oplus sl(2,\mathbb{C})$: (i) so(4) = su(2)⊕su(2): (0,6). (ii) so(2,2) = su(1,1)⊕su(1,1): (4,2). (iii) so*(4) = su(2)⊕su(1,1): (2,4). (iv) so(3,1) = $sl(2,\mathbb{C})_\mathbb{R}$ = Fix(σ_Z), $\sigma_Z(X_L,X_R)=(\bar{X}_R,\bar{X}_L)$ (the τ=conjugation instance of F30.2b), Killing eigenvalues $\{-4,-4,-4,+4,+4,+4\}$: (3,3). The swap (iv) is the unique factor-exchanging case and is, by Theorem F30.2b, the realification $sl(2,\mathbb{C})_\mathbb{R}$. [IMPORTED-PROVEN; numerics in Appendix C.]

Appendix B. The Steiner-centered rapidity, the composition theorem, and the holonomy — derivation

B.1 Steiner centering. The Steiner point $s(K) = (1/\pi)\int_0^{2\pi} h_K(\theta)\, n(\theta)\, d\theta$ picks out the n=1 mode: with $h = w/2 + \Sigma(a_n \cos n\theta + b_n \sin n\theta)$, $s = (a_1, b_1)$, since $\int\cos^2\theta\, d\theta = \pi$ and all other modes integrate to zero against $(\cos\theta, \sin\theta)$. It is translation-covariant: $h_{K+a} = h_K + a\cdot n$ adds a to the n=1 mode, so $s(K+a)=s(K)+a$, and $\tilde{h} = h - s\cdot n$ is translation-invariant. For a constant-width curve all even modes vanish (F30.2a); removing the n=1 mode by centering leaves only odd $n \geq 3$, and $\rho_K = ½ \ln(\tilde{h}_1/\tilde{h}_2)$ is a translation-gauge invariant. The leading model $h = w/2 + (w/16)\cos 3\theta$ has $a_1=b_1=0$, so its Steiner point is the origin and it is already centered.

B.2 ρ = artanh β. Put $\beta = (h_1-h_2)/(h_1+h_2) = (h_1-h_2)/w$. Positivity of h_1,h_2 gives $|h_1-h_2| < h_1+h_2$, so $|\beta| < 1$. Then $1+\beta = 2h_1/w$, $1-\beta = 2h_2/w$, so $h_1/h_2 = (1+\beta)/(1-\beta)$ and $\rho = ½ \ln(h_1/h_2) = ½ \ln[(1+\beta)/(1-\beta)]$ = artanh β. Since the seam exchanges $h_1 \leftrightarrow h_2$, $\beta(\theta+\pi) = -\beta(\theta)$ and $\rho(\theta+\pi) = -\rho(\theta)$. For the leading model $\beta = (\cos 3\theta)/8$, $\max|\beta| = 1/8$, and ρ_{max} = artanh(1/8) = ½ ln(9/7) ≈ 0.1257.

B.3 Lorentz brackets. With $\mathbf{J}=-(i/2)\sigma$, $\mathbf{K}=(1/2)\sigma$ and $[\sigma_i,\sigma_j]=2i\varepsilon_{ijk}\, \sigma_k$: $[J_i,J_j]=\varepsilon_{ijk} J_k$, $[J_i,K_j]=\varepsilon_{ijk} K_k$, $[K_i,K_j]=-\varepsilon_{ijk} J_k$; **K** Hermitian, **J** anti-Hermitian (check A1). $G_L = \exp[½(\rho+i\varphi)n\cdot\sigma] = B\cdot R$ with $B=\exp(½\rho\, n\cdot\sigma)$ Hermitian, $R=\exp(½i\varphi\, n\cdot\sigma)$ unitary; det $G_L=1$; unitary iff $\rho=0$. Seam: $\rho\to-\rho \Rightarrow B(\theta+\pi)=B(\theta)^{-1}$.

B.4 Composition (F30.7). $\rho(\theta)$ is continuous, seam-odd, not identically zero, so its image is $[-\rho_{max}, \rho_{max}] \ni 0$ (an open neighborhood of 0). For $\xi \in \mathbb{R}$ pick $N \geq \lceil|\xi|/\rho_{max}\rceil$ and θ_N with $\rho(\theta_N)=\xi/N$ (IVT). Same-axis boosts commute, so $B(\theta_N)^N = \exp(½\xi\, n\cdot\sigma)$. Every $g \in SL(2,\mathbb{C}) = U\cdot P$ (polar), $P=\exp(½\xi\, n\cdot\sigma)$, $U \in SU(2)$; KAK then gives $g = U_1 \exp(½\xi\, n\cdot\sigma)\, U_2$. So elementary support-boosts and rotations generate $SL(2,\mathbb{C})$ (check C5: composition to 2×10^{-11}, KAK to 4×10^{-15}).

B.5 Causal cone. $X = x_0\mathbb{1} + x\cdot\sigma$: det $X = x_0^2-|x|^2$; eigenvalues $x_0\pm|x|$; $X\geq0 \Leftrightarrow x_0\geq|x|$. A rank-one positive Hermitian matrix equals $\psi\psi^\dagger$ with $\det(\psi\psi^\dagger)=|\psi_1|^2|\psi_2|^2-|\psi_1\bar{\psi}_2|^2=0$ and $\mathrm{tr}(\psi\psi^\dagger)=|\psi|^2>0$: the future null cone is the spinor outer products (check D1). For $g \in SL(2,\mathbb{C})$, $g\psi\psi^\dagger g^\dagger = (g\psi)(g\psi)^\dagger$ is again a future null outer product (cone preserved, orthochronous: check D3); on the interior $gXg^\dagger$ stays positive definite (check D2).

Appendix C. Numerical verification

SYMBOLIC-THEOREM. (A1) [J,J]=J, [J,K]=K, [K,K]=−J exact; **K** Hermitian, **J** anti-Hermitian. (A2) $\det X = x_0^2-x_1^2-x_2^2-x_3^2$ symbolically. (A3) cos n(θ+π) and sin n(θ+π) both = ±(odd/even) ⇒ constant width ⇔ even cosine and sine modes vanish, J = −I on odd. (A4) Doubled-Simple τ-form σ^2=id for random τ-inputs; Fix Killing (3,3) = so(3,1). GEOMETRY. (B1) ρ W-independent to 3×10^{-16}, seam-odd to 6×10^{-16}. (B2) centered circle ρ=0; asymmetric curve ρ≠0. (B3) $h+h'' \geq 0$; J = −I (odd). (B4) raw ρ gauge-dependent (shifted circle 0.35); Steiner-centered ρ invariant (circle 2×10^{-16}, model 3×10^{-16}). (B5) ρ = artanh β to 10^{-9}, |β|<1; exact Reuleaux $a_3 = w/4\pi \approx 0.0796$, harmonics 3,9,15, range ±0.156. FULL-HOLONOMY. (C1) actual-generator real rank 6 (single angle 3). (C2) det G_L=1; unitary iff ρ=0. (C3) $B(\theta+\pi)=B(\theta)^{-1}$ to 10^{-7}; B Hermitian. (C4) $(X\_L, \bar{X}\_L) \in \mathrm{Fix}(\sigma\_Z)$. (C5) composition to 2×10^{-11}; KAK to 4×10^{-15} — G-6 closed. CAUSAL-CONE. (D1) $\psi\psi^\dagger$ Hermitian, det=0, trace>0, PSD over 6000 spinors. (D2) interior future timelike over 6000 points. (D3) $SL(2,\mathbb{C})$ orthochronous over 3000 elements. (D4) chain → signature (1,3). DYNAMICS-DIAGNOSTIC. (E1) z* = 0.43828+0.36059i, modulus 0.89151; $K f_+ K^{-1} = f_-$. (E2) modulus ≠ 1 and < 1. (E3) $T^2 = -1$ for j=½. REGRESSION. (F1) A=35/437; (F2) Q=11; (F3) (Z,X,Y)=(2,3,6); (F4) four signatures distinct; (F5) factors commute. SCOPE-AUDIT. (G1) G-6 closed; (G2) G-7 identification; (G3) physical-T OPEN; (G4) temporal POVM OPEN; (G5) circle-selection and Reuleaux-extremality OPEN, A29 untouched. All categories consistent.

References

Z-Spin corpus

[ZS-F2] K. Kang, "Geometric Impedance A = 35/437," ZS-F2 v1.0 (2026).
[ZS-F4] K. Kang, "The Z_2 Seam and SU(2) Double Cover," ZS-F4 v1.0 (2026).
[ZS-F5] K. Kang, "Gauge Symmetry Constraint: Why Q = 11, and the Seam Involution Ŝ," ZS-F5 v1.0 (2026).
[ZS-F7] K. Kang, "Twin-Reuleaux Holonomy, the J-Involution, and Constant-Width Pair Kinematics (§6, §11)," ZS-F7 v1.0(Revised) (2026).
[ZS-F11] K. Kang, "The Operational Observer Coordinate," ZS-F11 v1.0 (2026).
[ZS-F14] K. Kang, "Z-Anchored Vortex and the Twin-Reuleaux Joint ODE," ZS-F14 v1.0 (2026).
[ZS-F18] K. Kang, "The Three-Bridge Meta-Map and Three-Dimensional Self-Organization," ZS-F18 v2.1 (2026).
[ZS-M1] K. Kang, "i-Tetration and the Fixed Point," ZS-M1 v1.0 (2026).
[ZS-M2] K. Kang, "Complexified Lorentz Algebra, the Chiral sl(2) Split, and so(4) ≅ sl(2)_L × sl(2)_R," ZS-M2 v1.0 (2026).
[ZS-M3] K. Kang, "Quantum Tetrahedron, j = 1/2 Uniqueness, and 4π Closure," ZS-M3 v1.0 (2026).
[ZS-M17] K. Kang, "Z_2 Seam Involution and Real-Form Selection (§17.5)," ZS-M17 v1.0 (2026).
[ZS-Q1] K. Kang, "Geometric Decoherence and the Born Rule," ZS-Q1 v1.0 (2026).

[ZS-Q14] K. Kang, "Kramers Structure and χ_Z = −1," ZS-Q14 v1.0 (2026).
[ZS-S15] K. Kang, "The Poynting–Commutator Theorem and Twin-Reuleaux Realization," ZS-S15 v1.0 (2026).
[ZS-A28] K. Kang, "JZ-Odd Doublet, Kramers Structure, and the Susceptibility Kernel," ZS-A28 v1.9 (2026).
[ZS-A29] K. Kang, "Rank-Weighted Vacuum Budget and the Coincidence Problem," ZS-A29 v1.6 (2026).

External

[1] É. Cartan, "Les groupes réels simples finis et continus," Ann. Sci. Éc. Norm. Supér. 31, 263 (1914); A. W. Knapp, *Lie Groups Beyond an Introduction*, 2nd ed. (Birkhäuser, 2002), Ch. VI (real forms of so(4,ℂ): so(4), so(3,1), so(2,2), so*(4)).
[2] R. Penrose and W. Rindler, *Spinors and Space-Time*, Vol. 1 (Cambridge Univ. Press, 1984), §1.2–§1.4 (Hermitian X, det X = Minkowski norm, future null cone {ψψ†}, SL(2,ℂ) → $SO^{+}(1,3)$ orthochronous).
[3] T. Bonnesen and W. Fenchel, *Theory of Convex Bodies* (BCS Associates, 1987); H. Groemer, *Geometric Applications of Fourier Series and Spherical Harmonics* (Cambridge Univ. Press, 1996), Ch. 4 (support functions, the Steiner point as the first Fourier coefficient, constant width as the vanishing of even harmonics).
[4] M. Tegmark, "On the dimensionality of spacetime," Class. Quantum Grav. 14, L69 (1997).
[5] I. Bars, "Survey of two-time physics," Class. Quantum Grav. 18, 3113 (2001).
[6] H. van Dam and Y. J. Ng, "Why 3+1 metric rather than 4+0 or 2+2?," Phys. Lett. B 520, 159 (2001).
[7] H. A. Kramers, Proc. Amsterdam Acad. 33, 959 (1930) (T^2 = (−1)^{2j}).
[8] A. S. Holevo, *Probabilistic and Statistical Aspects of Quantum Theory* (North-Holland, 1982) (covariant time observables).

Version History

v1.0 (June 2026): Initial release; claimed signature closure and a temporal-Born no-go; single 24/24 PASS ledger.
v1.1 (June 2026): Retracted 'closes the only axis'; cited M2/M17.5; added a three-real-form forcing-and-exclusion with so(2,2); formalized T=U_T K, T^2=−1; retracted the temporal-Born no-go to OPEN; reclassified the ledger; added the Open Gates Registry.
v1.2 (June 2026): Mathematical-core revision. Completed the classification to four real forms (so*(4), (2,4)); promoted the swap to a constant-width lemma; added a matrix holonomy lift (compact SU(2), boost by 'multiply by i'); replaced the circular tetrad with the η-free det X = Minkowski reconstruction; positioned against Tegmark/Bars/van Dam–Ng; separated σ_Z from physical T; restructured the verification.
v1.3 (June 2026): Chain-closure revision. Introduced the support-ratio rapidity $\rho=\frac{1}{2}\ln(h_1/h_2)$; lifted the holonomy to the full six-dimensional so(3,1) with [K,K]=−J and the boost-inversion identity; added the Doubled-Simple Exchange and constant-width Fourier theorems; extended det X = Minkowski to a causal-cone reconstruction; assembled the chain; registered gates G-6, G-7.

v1.4 (June 2026): Gauge-invariance and honesty revision. (1) Made the support-ratio rapidity **translation-gauge invariant** by **Steiner centering** $\rho\_K = ½ \ln(\tilde{h}_1/\tilde{h}_2)$, $\tilde{h} = h - s(K)\cdot n$, repairing the v1.3 origin-dependence (a translated circle no longer shows a spurious boost) (§5.1–5.2, F30.3a). (2) Justified the log-ratio canonically: **ρ = artanh β** with $\beta=(h_1-h_2)/w$, $|\beta|<1$ — the special-relativistic rapidity–velocity relation (§5.3). (3) Separated **three layers** — general constant-width / exact Reuleaux (harmonics 3,9,15, leading $w/4\pi$, range ±0.156) / corpus leading model ($w/16$, range ±0.126) — correcting the v1.3 "entire content is $(w/16)\cos 3\theta$" (§5.4). (4) Added the **Rapidity-Composition Theorem** (F30.7), generating the whole $SL(2,\mathbb{C})$ identity component from bounded elementary support-boosts plus $SU(2)$ rotations and **closing G-6** (§7.2). (5) Recast the **Doubled-Simple Exchange Theorem in τ-form**, σ^2=id automatic (§4.2, F30.2b). (6) Tested the **actual** holonomy generators (rank 6 from two angles, rank 3 from one) rather than an independent basis (§5.5, C1). (7) Reduced overclaim: F30.5 marked **DERIVED-CONDITIONAL** with the chain's physical step isolated as G-7, the title softened to "a conditional … construction," and the circle/Reuleaux dichotomy demoted to a safe half plus OPEN G-8 (§5.7). (8) Restructured the verification into the same seven categories and renamed it zs_f30_verify_v1_4.py. Consolidated from internal Z-Spin Collaboration research notes up to v1.4.0.